Optimizing tour plan using Sentimental Analysis

Abstract

Numerous tourist attractions along with a huge amount of information about them on web and social platforms have made the decision-making process for selecting and visiting them complicated. This paper introduces a tourism recommendations system that uses sentimental analysis of the user to provide personalized recommendations. By asking certain questions without asking explicitly, we will analyze the mood of the user. For analyzing the mood of the user, we will be using VADER for sentimental analysis. Based on the mood of the user, it will recommend different places. In addition, the system utilizes the vital context-aware recommendations like location, weather, time, to filter unsuitable items and increase the quality of suggestions regarding the current situation.

Keywords: Sentimental analysis, VADER, context-aware

Introduction

With the expansion of tourism websites and social networks, a huge number of reviews and comments are produced and posted regularly. These travel websites use these reviews to suggest places to the tourists. As user's reviews are considered for suggesting places, personalized recommendations are missing in many of them. However, it is a challenge to suggest the best places for traveling to the user based on their preferences. As a travel recommendation system, it is important to recommend places based on the user's preference and not suggest the same travel plan to everyone. So, to overcome this drawback of the existing systems, sentiment analysis is an important factor to consider for personalized results for recommending places.

It is noteworthy that in the existing type of tourism recommendation systems, the frequent keywords used in the comments are exploited, regardless of the sentiments of users. As a result, the negative words that are stressed in the user's text might be mistakenly returned as their preferences. While sentiment analysis is very important in the tourism domain, it has been overlooked by existing travel planners in most cases. In fact, a comprehensive recommendation system in the context of tourism should include the following features:

- It should suggest places based on the user's mood(sentiment) instead of just using previous reviews or comments.
- Personalized recommendations should be used rather than suggesting the preplanned(same) plan to everyone.
- Day to day travel plans should be provided to the user to visit and enjoy every recommended tourist spot in a day.

To the best of our knowledge, none of the available tourism recommendation systems meets all the above features. Hence, this paper proposes context-aware tourism by sentiment analysis. In this system, sentimental analysis is leveraged to extract the user's preferences precisely. Along with that, context-aware recommendation systems are also included for perceiving user’s preferences successfully.

To address this need of providing personalized results to travelers, tailored to their preferences and emotions, this paper proposes a novel mood-based tourist recommendation system that integrates sentiment analysis, natural language processing, and multi-criteria decision making. By analyzing user responses to mood-related questions, our system predicts the user's current mood and suggests tourist destinations accordingly. Additionally, the system considers factors such as travel distance, budget, and hotel preferences to provide a comprehensive and optimized recommendation. Now, once the user finalizes the destination, the system provides them with different clusters of tourist spots in that destination that can be visited in a single day, based on the number of days of tour provided by the user.

This paper presents the design and development of the proposed system, highlighting its potential to enhance the travel experience and provide personalized recommendations to tourists.

Related work

Most of the tourism recommendation systems are based on user reviews in social media or the ratings from the user. The system collects reviews from other tourists about various attractions to determine the features and qualities of those places. Finally, the system combines the user's preferences with the characteristics of tourist attractions. It also considers the current context, such as weather, time, location, and user preferences, to make recommendations. However, the drawback of these methods is that this idea does not consider the number of people going for traveling. Simply suggesting the best places based on the reviews and previous history and context aware (location, time, weather, mood) does not suggest the best path or hotel for traveling.

Another tourist recommendation system “Personalized Travel Recommendation Based on Sentiment-Aware Multimodal Topic Model” uses UGC (User Generated content) like Textual comments, Reviews, images uploaded by users on existing travel websites like TripAdvisor and trip.com. From this UGC they analyze the sentiment of that particular user and then recommend tourist places to them. As they are using user’s social media data (maybe a travel website) to recommend places, there is a chance for users who have never used the system before or for newly added tourist attractions, there is little to no data available, making it hard to provide recommendations.

One of the popular travel companies MakeMyTrip (MMT) recommends places using user preferences, search history, location-based suggestions, user reviews, ratings, trending destinations, and budget-friendly packages. It tracks past behavior to offer personalized travel options, highlighting nearby attractions, popular hotels, and seasonal deals. However, MMT has limitations such as an

over-reliance on user ratings, which can be biased or outdated. Its personalization is limited and might not fully consider individual user preferences, often ignoring specific travel interests or sentiments. It focuses more on geographic proximity and popular destinations, overlooking niche options. Additionally, MMT's static recommendations fail to consider real-time factors like weather or events, which can affect travel plans.

Considering users' sentiments is vital in extracting their preferences from their reviews to avoid identifying negative points as preferences. To this end, this paper proposes a tourism recommendation system that exploits sentiment analysis to identify user preferences as well as key features of tourist attractions. The proposed method improves earlier research by including sentiment analysis (understanding emotions) in the analysis of tourist reviews using text mining. Unlike the studies it builds on, this new system is context-aware, meaning it also takes into account various factors like time, location, weather, user preferences, and the characteristics of tourist attractions when analyzing the reviews. This makes the system more tailored to each situation and user.

Novelty

Utilizing reviews and comments for recommendation can sometimes result in misguided suggestions. For a better recommendation system, it is wise to analyze the real time sentiment of the user to recommend places. Therefore, our system (by using Sentimental Analysis) will interact with the user in real time to know about their mood and based on this mood different places will be recommended. One of the major advantages of our system is that it gives personalized recommendations to everyone based on their mood, budget and how far they wish to travel. By utilizing these key factors this system will suggest a day-to-day tour plan for the recommended tourist spots, best hotel to stay to optimize users' happiness in the limited budget and by making cluster-based tourism spots, the user can travel multiple places without worrying about the expense. This way, we will improve the existing travel system to enhance the tourist's experience.

Proposed solution

At first, this website will interact with the user to know about their current mood. It will ask 7-10 questions with multiple options to choose from. Based on the user's choices the Sentimental Analysis tool (VADER) will generate a sentiment score which can be used to detect mood (Adventurous, Relaxed, Romantic, Happy, Depressed, Sad). It will also ask different travel related questions like what is their budget, how many days they want to travel, how far they can travel, etc.
The questions will come dynamically i.e. the user's choices will impact the upcoming questions. Therefore, two different users may get different sets of questions.

Sentiment Score Range	Mapped Mood
-1.0 to -0.6	Depressed
-0.6 to -0.3	Sad
-0.3 to 0.0	Neutral to Sad
0.0	Neutral
0.1 to 0.3	Relaxed
0.3 to 0.5	Romantic
0.5 to 0.7	Adventurous
0.7 to 1.0	Happy

Note: The user can skip this part as well if they know where they want to go and directly go to the Tour Plan Page.

Example Questions:
How do you feel about spending time with friends right now?

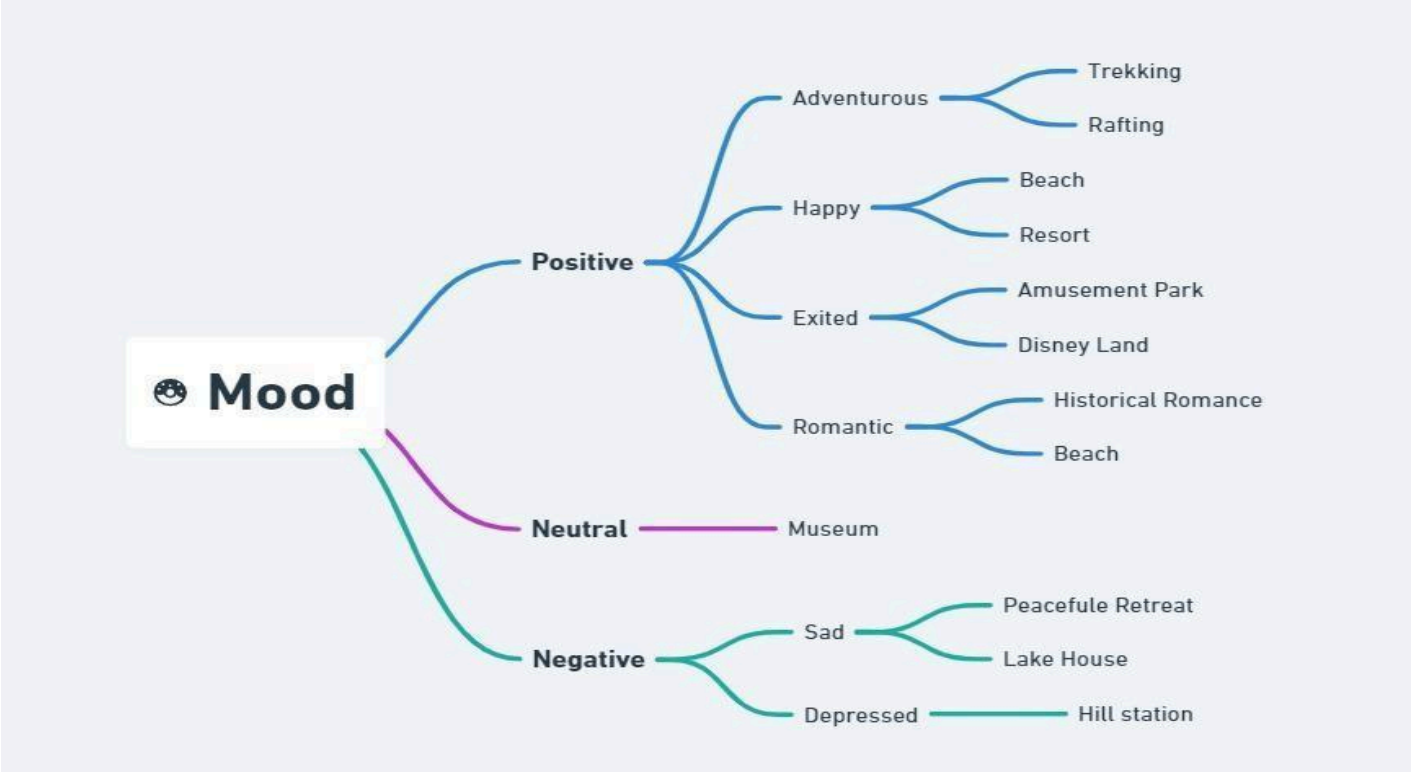
1. "I’m looking forward to having a great time with them!" (Happy, Enthusiastic)
2. "I’m in the mood for a quiet, intimate evening with someone close." (Romantic, Relaxed)
3. "I’d rather be alone today." (Sad, Withdrawn)
4. "I’m feeling too stressed to be around people right now." (Stressful)

What would you like to do for yourself right now?

1. "Do something spontaneous and exciting!" (Adventurous, Enthusiastic)
2. "Relax with a book or a bath." (Relaxed)
3. "Spend time with someone special." (Romantic)
4. "Just take a nap or sleep it off." (Sad, Tired)
5. "I'm feeling too stressed to focus on self-care." (Stressful)

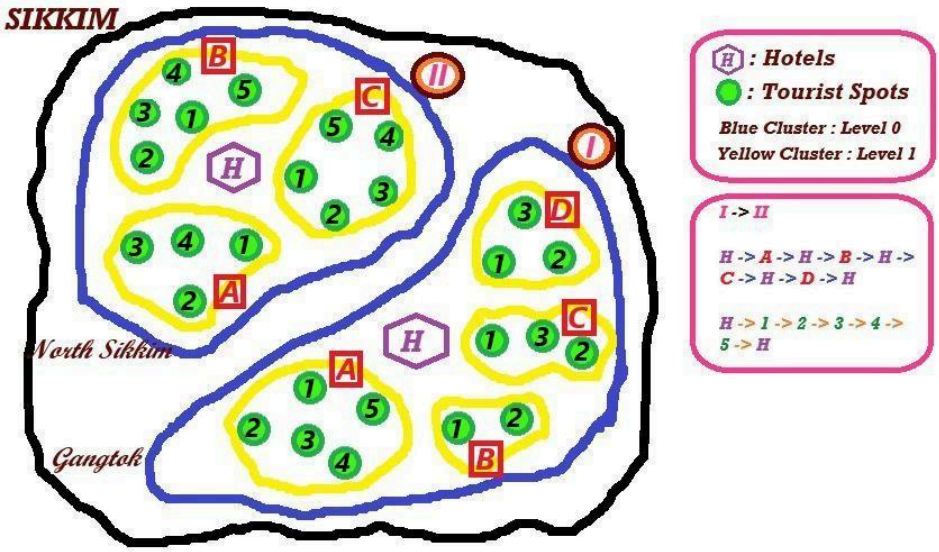
Based on the mood and other questions the model will recommend multiple destinations to choose from. Our Recommendation model will have **mood-destination** type mapping to recommend a particular destination to a specific type of mood, for example A relaxed person will be recommended something like beaches or temples. The recommendation model will consider other factors like budget, number of days as well.

Mood	Tourist Place Suggestions
Depressed	Quiet retreats, wellness resorts, spiritual places (e.g., Rishikesh, Auroville, Dharamshala)
Sad	Scenic nature spots, calm beaches, hill stations (e.g., Coorg, Munnar, Alleppey)
Neutral	Urban cities with mild activities (e.g., Pune, Ahmedabad, Hyderabad)
Relaxed	Beaches, countryside stays, spas (e.g., Goa, Pondicherry, Kerala backwaters)
Romantic	Romantic destinations, lakes, and cozy resorts (e.g., Udaipur, Shimla, Jaipur)
Adventurous	Trekking spots, adventure sports hubs (e.g., Manali, Rishikesh, Ladakh)
Happy	Party spots, vibrant cities, cultural festivals (e.g., Mumbai, Varanasi, Rajasthan)



After selecting a destination from multiple recommendations, we will use the concept of **Skyline Computation** to recommend hotels (best to worst) based on 3 criteria: Budget, Proximity to Tourist destinations and star rating. All 3 criteria will have different weightage for each user. The weightage will depend on users' choices during those questions at start, for example who prefers luxury will have less weightage for budget but more for proximity to tourist spots and star rating.

We will use cluster of clusters concept, where the whole destination (Example: state of Sikkim) will be divided into multiple clusters(level-0), such that each cluster can be traveled by staying in a single hotel (Example: Gangtok, Darjeeling, and North Sikkim can be three clusters). Inside each level-0 cluster there will be multiple clusters(level-1) as well. This cluster will contain all the sights which can be traveled in a single day. So, for example, a 3-day tour plan might contain three level-1 clusters (Each cluster for a day) and one level -0 cluster. In case of long tours like more than 7 days, there can be multiple level-0 clusters, therefore the user needs to stay in multiple hotels. Users will be recommended hotels in each level-0 cluster. We will use Taxicab Analysis to form these level-1 clusters as we must keep in mind the modular distance between the coordinates of the spots instead of Euclidean distance. We will use the Total Ordering concept for recommending the sequence of traveling to all those hotspots.



This tour plan will purely be a recommendation as users can modify it as well. They can change the order, remove, or add tourist spots as per their needs.

So, here is a case study on how this model works –

As sentiment analysis is done by using the Natural Language Toolkit (NLTK), an open-source library for Natural Language Processing (NLP) in python; so, at first the required libraries are imported and the packages of NLTK are downloaded. In the dataset, each city is associated with a description (Fig. 1.1). Now, keywords from the descriptions are obtained, and then these keywords are mapped with several categories (Fig. 1.2) (For ex: "Beaches": ["beach", "coastal", "ocean", "sea", "island"]; here the keywords: beach, coastal, ocean, sea, and island, are mapped into the category of beaches). Now, each city is provided some category (it can be multiple also) based on the keywords obtained from their description (Fig. 1.3). Next, the categories are mapped with the moods (Fig. 1.4) (For ex: 'Excited': ['Amusement Parks', 'Festivals', 'Music Concerts']; here the categories: amusement parks, festivals, music concerts, are mapped with the mood: excited). Now, each city is assigned with all the possible moods, fetched from the mood-category mapping, corresponding to the categories that the city has

(Fig. 1.5). Now, each mood is mapped with a particular range of sentiment scores (Fig. 1.6). With this, a recommendation is added for each mood.

Now, all the responses that are obtained from the mood-related questions asked to the user will be fed to the VADER sentiment analyzer (SentimentIntensityAnalyzer().polarity_scores(response)). This will provide the negative, neutral, positive, and compound (average of pos, neu, neg) scores for each response (Fig. 1.7). Now, by averaging the compound scores of all the responses, we will get the final sentiment score. Once the sentiment score is known, the mood will be easily predicted from the mood-sentiment_score mapping. And once the mood is predicted, the places will be now shortlisted, as the cities are mapped with the moods. In this way, the tourist spots will be suggested to the user based on their mood.

Alongside mood-related questions, questions regarding how far they can travel, how much money they can spend, their tour preferences, these will also be asked to the user and based on these answers, final places will be recommended to them. So, at first, we need to calculate the distance between the user's location to all the tourist spots which were suggested from the mood analysis (Fig. 2.1.1). To calculate the distance between any two places, we have integrated an API -> OpenRouteService API. This will generate the distance between any two places, provided the latitudes and longitudes of those two places. Now, based on the user's choice regarding how far they can travel, the results are filtered out (For example, whether they can travel upto 200m/upto 500m/anywhere) (Fig. 2.1.2). From this shortlisted result, the user must pick up one tourist spot. In this way, the tourist destination will be recommended to the user based on their current mood and preferences.

Once the user has finalized their destination from the list of places suggested to them, now, the model extracts the latitude and longitude of that place using Google Geocoding API. Thereafter, the location (latitude and longitude) of the city is fed to the Google Places API, which fetches all the tourist spots of that city, or near to it. Next, by using predefined python function (KMeans), the model groups the tourist spots close to each other and forms clusters that can be visited in a single day, based on the number of days of tour provided by the user (For example, if the selected destination is 'Puri' and the number of days of tour is '4', then the four clusters of tourists spots in Puri is shown in Fig. 3).

Fig. 2.1.1: Dist. from Howrah (User's location to different tourist places)

	City	Distance (km)
0	Goa	2083.83713
1	Rishikesh	1564.85013
2	Jaipur	1526.64650
3	Kolkata	6.02591
4	Jodhpur	1861.70638
5	Amritsar	1917.65163
6	Wayanad	2097.85000
7	Kochi	2268.04400
8	Ahmedabad	1952.18138
9	Mysore	1976.02538
10	Ujjain	1621.40813
11	Gwalior	1282.64263
12	Mathura	1350.04975
13	Vrindavan	1351.03988
14	Ranthambore	1524.45850
15	Ajmer	1659.11663
16	Puri	492.98166
17	Bikaner	1861.62225
18	Hogenakkal	1918.83850
19	Pushkar	1671.98425

Fig. 2.1.2: Filtered places according to user's choice (Anywhere)

Case Study

In the modern time we are planning to built personalized recommendations travel plans.our focuses on an AI-driven sentimental analysis system that suggest tourist spot or destination on the based on user's mood.The main idea to predict the user's emotional state based on their responds with set of predefined question we will recommended a tourist spot accordingly.

We are using "Indian Places to Visit Reviews Data" Dataset for our system to recommend tourist place ,Dataset which have column city,place,review,rating…so on its size is (1482466 x 7) i.e. the column contains only 7 column and 1482466 rows. The dataset has been pre-processed by taking only the important columns like city name and the reviews, so the unwanted columns have been removed like place, date, rating etc, and the dataset has been modified within 2 columns and row size of 1794(Fig. 1.1). Since the city column consists of multiple common cities names it has been compiled by removing the redundant cities. The processed dataset used in this project consists of only city name and combination of review and raw review.

	City	Review
0	Aamby Valley City	aamby valley beautiful place clear blue skies ...
1	Abhaneri	visit jaipur trip usually itinerary need ask t...
2	Acharapakkam	hill shrines situated tamil nadu history place...
3	Achooranam	wayanad tea museum awesome place wayanad miss ...
4	Achrol	say overwelhmed expecting world class service ...

Fig. 1.1: *Dataset*

The reviews of each city have been extracted and map to the given following categories based on beaches, hills, resorts etc(Fig. 1.3).

	City	Review	Categories
0	Aamby Valley City	aamby valley beautiful place clear blue skies ...	Nature Reserves, Mountain Expeditions, Seclude...
1	Abhaneri	visit jaipur trip usually itinerary need ask t...	Beaches, Spas, Nature Reserves, Quiet Resorts,...
2	Acharapakkam	hill shrines situated tamil nadu history place...	Art Galleries
3	Achooranam	wayanad tea museum awesome place wayanad miss ...	Spas, Hiking Trails, Vineyards, Nightlife Hots...
4	Achrol	say overwelhmed expecting world class service ...	Beaches, Spas, Nature Reserves, Hiking Trails,...

Fig. 1.3: *Categories assigned to cities*

After that we categorize the locations based on reviews from the updated Dataset. Using predefined keywords to

classify each location based on the description which developed a function to match location description with the closest category(Fig. 1.2). Here, we mapped these categories to the appropriate mood-based recommendation while applying this below method to enhance the relevance of recommendations(Fig. 1.4).

```
# Define categories and associated keywords
categories = {
    "Beaches": ["beach", "coastal", "ocean", "sea", "island"],
    "Spas": ["spa", "wellness", "relax", "retreat"],
    "Nature Reserves": ["nature reserve", "wildlife", "sanctuary", "park"],
    "Quiet Resorts": ["resort", "quiet", "peaceful", "secluded"],
    "Hiking Trails": ["hiking", "trail", "trek", "walk"],
    "Mountain Expeditions": ["mountain", "expedition", "climb", "peak"],
    "Scuba Diving": ["scuba", "diving", "underwater", "coral"],
    "Vineyards": ["vineyard", "wine", "grape", "tasting"],
    "Coastal Resorts": ["coastal", "beach", "resort", "ocean"],
    "Secluded Cabins": ["cabin", "secluded", "quiet", "forest"],
    "Urban Centers": ["city", "urban", "downtown", "metropolitan"],
    "Amusement Parks": ["amusement park", "rollercoaster", "theme park", "rides"],
    "Nightlife Hotspots": ["nightlife", "bar", "club", "party"],
    "Historic Sites": ["historic", "ancient", "ruins", "landmark"],
    "Art Galleries": ["gallery", "art", "exhibition", "museum"],
    "Nature Walks": ["nature walk", "trail", "path", "outdoors"],
    "Festivals": ["festival", "celebration", "event"],
    "Music Concerts": ["concert", "live music", "festival", "band"],
}
```

Fig. 1.2: *Categories - Keyword mapping*

```
# Define mood-to-spot mapping
mood_to_spots = {
    'Sad/Depressed': ['Beaches', 'Spas', 'Nature Reserves', 'Quiet Resorts'],
    'Angry/Frustrated': ['Hiking Trails', 'Mountain Expeditions', 'Scuba Diving'],
    'Happy': ['Vineyards', 'Coastal Resorts', 'Secluded Cabins'],
    'Low/Unhappy': ['Urban Centers', 'Amusement Parks', 'Nightlife Hotspots'],
    'Neutral': ['Historic Sites', 'Art Galleries', 'Nature Walks'],
    'Excited': ['Amusement Parks', 'Festivals', 'Music Concerts']
}
```

Fig. 1.4: Mood - Category mapping

Then Google Distance Matrix API is used to filter out locations based on max distance travel by the user.
Sentimental Analysis Tool Vader(Valence Aware Dictionary and Enstiment Reasoner) is used to predict a user's mood. The user will answer some indirect mood related questions related to the user's emotions, music preferences, and daily experiences. The users are allowed to select multiple-choice answers, which were then processed for sentiment analysis. Based on the user's choice, mood will be detected.

NLTK library is used to assess the user's sentiment because VADER is good for optimized and short texts and it is relevant to use.
So necessary Python libraries have been imported such as Pandas, Numpy, Matplotlib, Pyplot, Seaborn.
The VADER plays an important role in analysing the mood of the user to recommend a city based on the user's sentiment. From the Q&A, it gives a sentiment score that varies from -1(negative score) to +1(positive score)(Fig. 1.6).

```python
# Define mood categories based on the sentiment score range
mood_map = {
    (-1.0, -0.3): "Angry/Frustrated",
    (-0.3, -0.2): "Sad/Depressed",
    (-0.2, -0.1): "Low/Unhappy",
    (-0.1, 0.1): "Neutral",
    (0.1, 0.3): "Happy",
    (0.3, 1.0): "Excited"
}
```

Fig. 1.6: Sentiment score - mood mapping

```
{'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.4404}
{'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.4404}
{'neg': 0.0, 'neu': 0.385, 'pos': 0.615, 'compound': 0.4939}
{'neg': 0.0, 'neu': 0.0, 'pos': 1.0, 'compound': 0.34}
{'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}

Average Sentiment Score: 0.34
```

Fig. 1.7: Average sentiment score based on the responses given by the user

Once the mood has been determined, it is mapped to the city since each city is mapped to a category like beaches, hills etc and each category is mapped to a mood which is based on the research, thus mood is mapped to the city(Fig 1.5). Thus, by analysing the mood of the user, recommendations for travelling are suggested.

	City	Review	Categories	Mapped_Mood
0	Aamby Valley City	aamby valley beautiful place clear blue skies ...	Nature Reserves, Mountain Expeditions, Seclude...	Sad/Depressed, Angry/Frustrated, Happy, Low/Un...
1	Abhaneri	visit jaipur trip usually itinerary need ask t...	Beaches, Spas, Nature Reserves, Quiet Resorts,...	Sad/Depressed, Angry/Frustrated, Happy, Low/Un...
2	Acharapakkam	hill shrines situated tamil nadu history place...	Art Galleries	Neutral
3	Achooranam	wayanad tea museum awesome place wayanad miss ...	Spas, Hiking Trails, Vineyards, Nightlife Hots...	Sad/Depressed, Angry/Frustrated, Happy, Low/Un...
4	Achrol	say overwelhmed expecting world class service ...	Beaches, Spas, Nature Reserves, Hiking Trails,...	Sad/Depressed, Angry/Frustrated, Low/Unhappy, ...

Fig. 1.5: Mood assigned to cities

- Google Places API: This API is used to fetch all the tourist places within a city.
- Geocoding API: This API is used to get the longitude or latitude of any tourist place or location.
- *"Kmeans"* is imported from "sklearn.cluster" which is used to cluster all the tourist places based on

proximity to each other.

User will choose one location from the multiple options recommended (based on their mood), the user's choice of city is passed to the Google Places API, which will fetch us all the tourist places in that location. Users will enter the number of days they want to travel.

The Geocoding API will help us to get the Longitude and Latitude of all the tourist places of that city.

Based on the longitude and latitude, the tourist places will be clustered with the help of K-Means clustering such that the tourist places which are near to each other are clustered together and one cluster can be travelled in a single day. Total number of clusters will depend on the number of days the user wants to travel.

```
Day 1:
- Bedi Hanuman Temple (RR3V+8R3, near bedi Hanuman, Chakra Tirtha Road, Puri)
- Sonar Gouranga Temple (Badasirei, Puri)
- ହରଚଣ୍ଡୀ ମନ୍ଦିର Harchandi Temple (RV32+WV7, Bhudan Nagar, Penthakata, Puri)

Day 2:
- Aruna Stambha (RR39+VJC, Puri)
- Shree Jagannatha Temple Puri (Puri)
- Light & Sound Show, Narendrakona, Puri (RR7G+8FJ, Badshahi Road, Puri)
- Shree Mandir Parikarma Prakalpa (RR6G+H23, Athar Nala Road, Balagandi, Puri)
- Puri (Market chack, near Anapurna Restaurant, Puri)

Day 3:
- Harachandi Temple, ହରଚଣ୍ଡୀ ମନ୍ଦିର (Sitala Pinda Sahi Road, Sandhajaga, Puri)
- Puri Tourism- Hotel, Taxi, Food (Heragouri, Puri)

Day 4:
- Shree Gundicha Temple, Puri (Gundicha Temple, Puri)
- Golden Sea Beach odisa (RR7J+9FQ, Puri)
- Chilka lake odisa (on the east coast of India, Puri, Khordha and Ganjam districts)
- Blue flag beach at Puri (RR7J+9FQ, Puri)
- Tourism Travells (Puri)
- KalingaTraditions (RR7J+9FQ, Puri)
```

Fig. 3: Clusters of tourist spots formed based on number of days of tour

Conclusion

In conclusion, this tourism recommendation system leverages sentiment analysis to provide a personalized and context-aware travel experience for users. By analyzing the user's mood through VADER and integrating key factors such as budget, travel duration, preferences, location, time, and

weather, the system offers tailored recommendations that align closely with the user's current emotional state. Unlike existing systems, our approach offers a comprehensive and adaptable travel plan that not only suggests suitable tourist destinations but also proposes the best accommodations and itineraries. The use of mood-destination mapping, along with skyline computation for hotel selection, ensures that users receive a seamless, enjoyable, and optimized travel experience that caters to their preferences. This innovative approach has the potential to revolutionize the tourism industry by enhancing user satisfaction and providing truly personalized travel experiences.